Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology

School of Computing

Department of Computer Science and Engineering

10211CS224 / FULL STACK APPLICATION DEVELOPMENT (JAVA)

PROJECT REVIEW : MILESTONE PROJECT – 1

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Real-Time Event Synchronization Engine

SDG 9 – Industry, Innovation and Infrastructure
(Best match for real-time systems & technology)

Presented By:

S SARAVANA ASWIN – VTU24809 – 23UECS0513.

Courses Handling Faculty:

Dr. Hafizur Rahman –

Assistant professor

INTRODUCTION

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- Real-time systems are important in modern applications.
- Many existing systems do not update data instantly.
- This causes delay and inconsistency for users.
- Users may see outdated or incorrect information.
- Real-time synchronization solves this problem.
- It ensures data updates instantly across all users.
- Used in chat apps, live tracking, and online systems.
- This project builds an efficient synchronization engine.

PROBLEM STATEMENT

- Existing systems rely on manual refresh.
- Data updates are slow and inefficient.
- Users experience inconsistent data.
- No proper real-time update mechanism.
- Delay affects performance of applications.
- Difficult to handle multiple users simultaneously.
- Leads to poor user experience.
- Need for a fast and reliable real-time system.

OBJECTIVE(S)

- To design and develop a real-time event synchronization system for efficient data handling.
- To implement instant data updates across multiple users without requiring manual refresh.
- To reduce latency and improve the overall responsiveness of the system.
- To maintain data consistency and accuracy across all connected clients in real time.

PROPOSED SYSTEM

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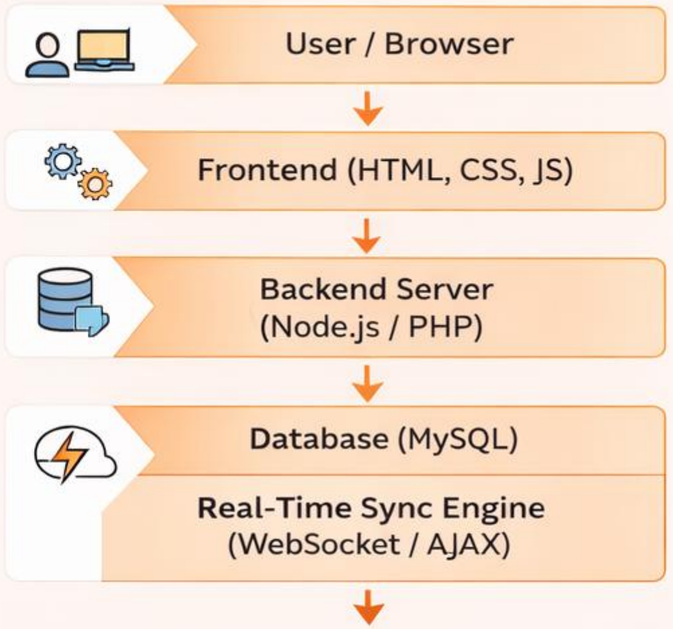
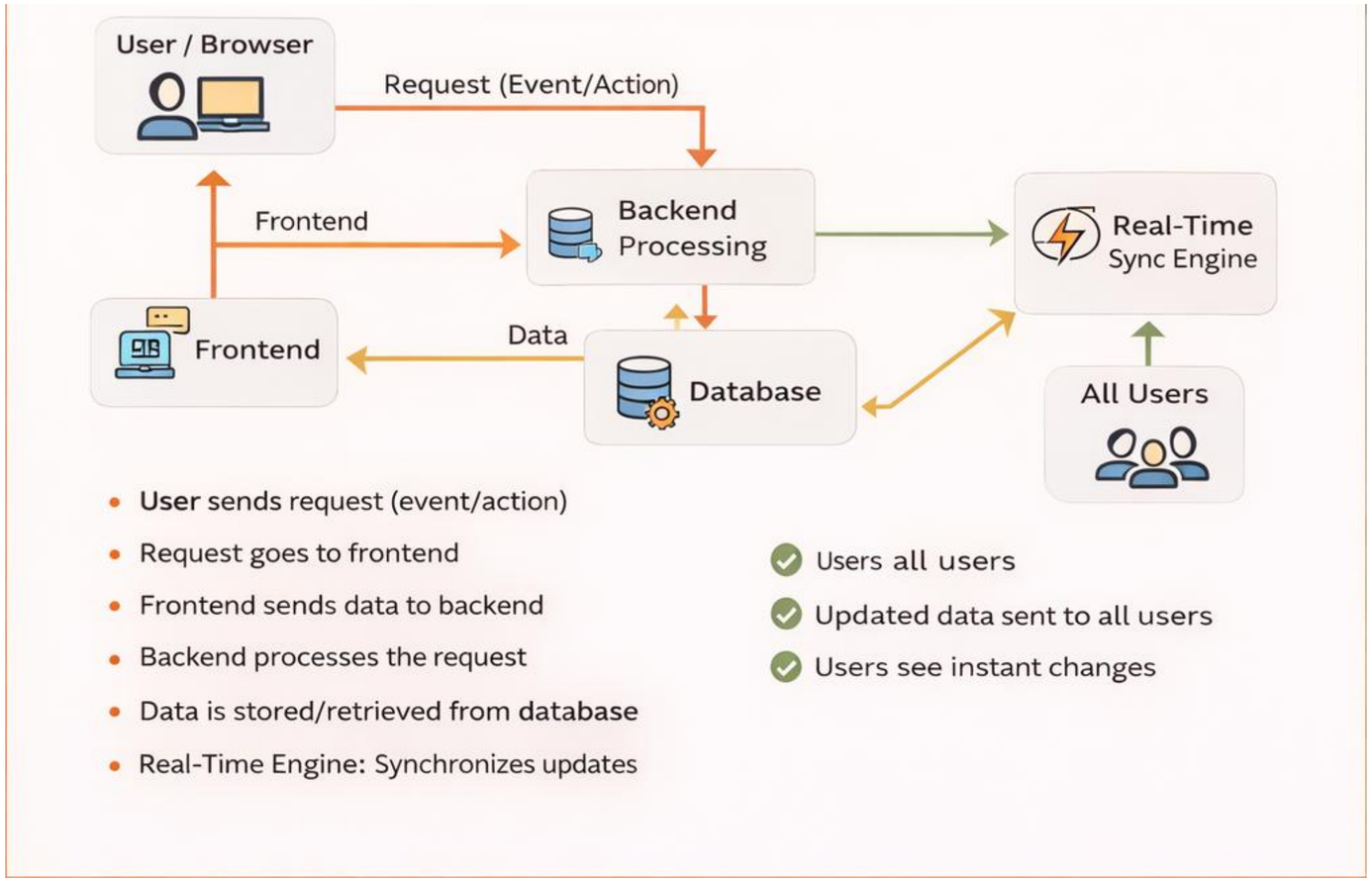
- Uses real-time communication techniques.
- Automatically updates data without refresh.
- Synchronizes data across all users instantly.
- Reduces delay and improves performance.
- Provides better user experience.
- Supports multiple users at the same time.
- Ensures data accuracy and consistency.
- Scalable and efficient system.

PROPOSED SYSTEM ARCHITECTURE



- **Input:** User actions (clicks, form data)
- **Processing:** Backend handles requests and logic
- **Database:** Stores and retrieves data (MySQL)
- **Output:** Updated data displayed to users instantly
- **Duput:** Updated data drsiplayed to users instantly
- **Real-Time Engine:** Synchronizes updates across all users

WORKFLOW


User / Browser
Request (Event/Action)
Frontend
Backend Processing
Real-Time Sync Engine
Frontend
Data
Database
All Users
User sends request (event/action)
Request goes to frontend
Frontend sends data to backend
Backend processes the request
Data is stored/retrieved from database
Real-Time Engine: Synchronizes updates
Users all users
Updated data sent to all users
Users see instant changes

TECHNOLOGIES USED

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Frontend:

- HTML
- CSS
- JavaScript

Backend:

- Node.js/PHP

Database:

- MySQL

Tools:

- VS Code
- XAMPP/Tomcat
- Browser (Chrome)

Hardware/Software Requirements:

- System with 4GB Ram or above
- Internet Connection
- Modern web browser

MODULE DESCRIPTION

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Module 1: User Management
Handles user login and authentication using username and password.
Ensures secure access based on user roles like admin, editor, and viewer.
Module 2: Core Processing
Implements the main system logic and processes user inputs efficiently. Handles event creation, validation, and execution within the system..
Module 3: Database Management
Manages storage and retrieval of user and event data using MySQL. Ensures data integrity, consistency, and efficient access.
Module 4: Reporting & Analytics
Provides insights through real-time dashboard statistics and logs. Displays event data for monitoring system performance and activity.

IMPLEMENTATION SCREENSHOTS

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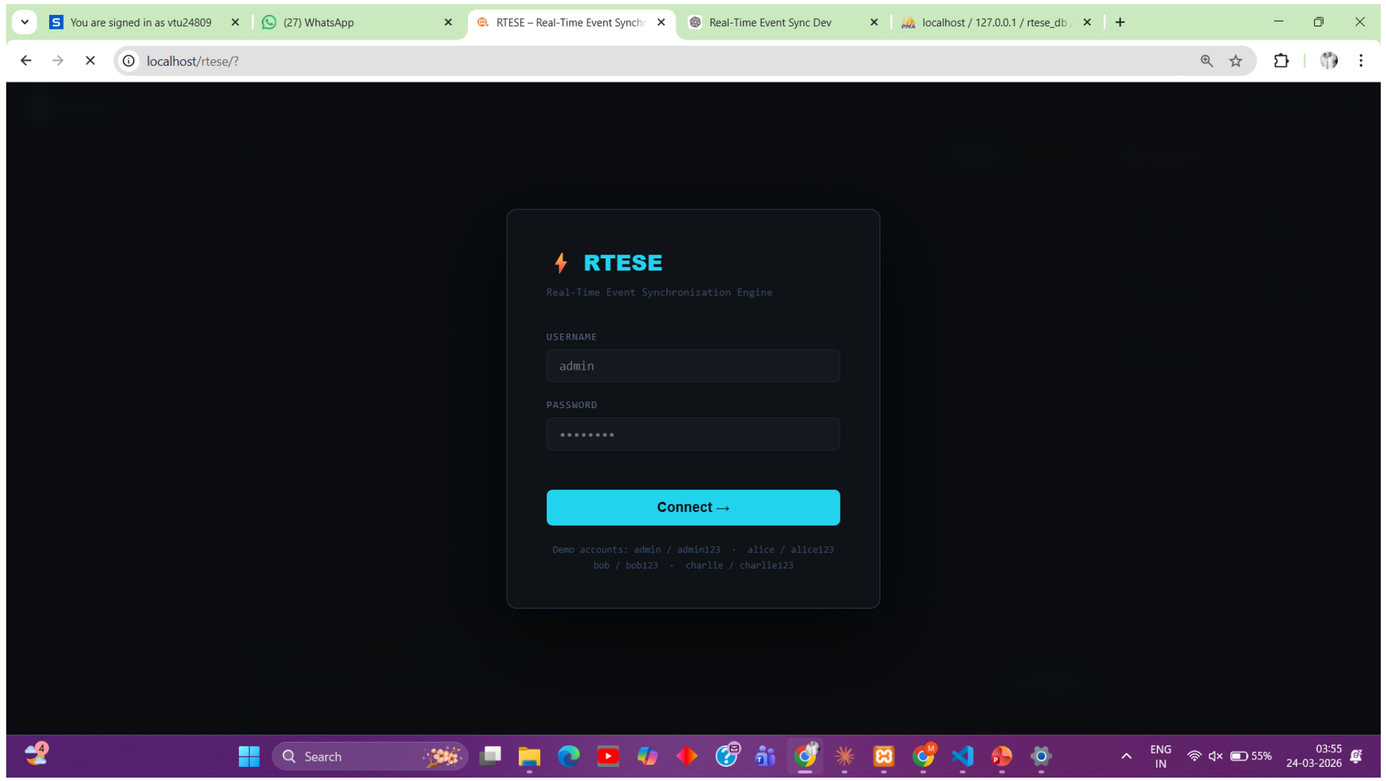
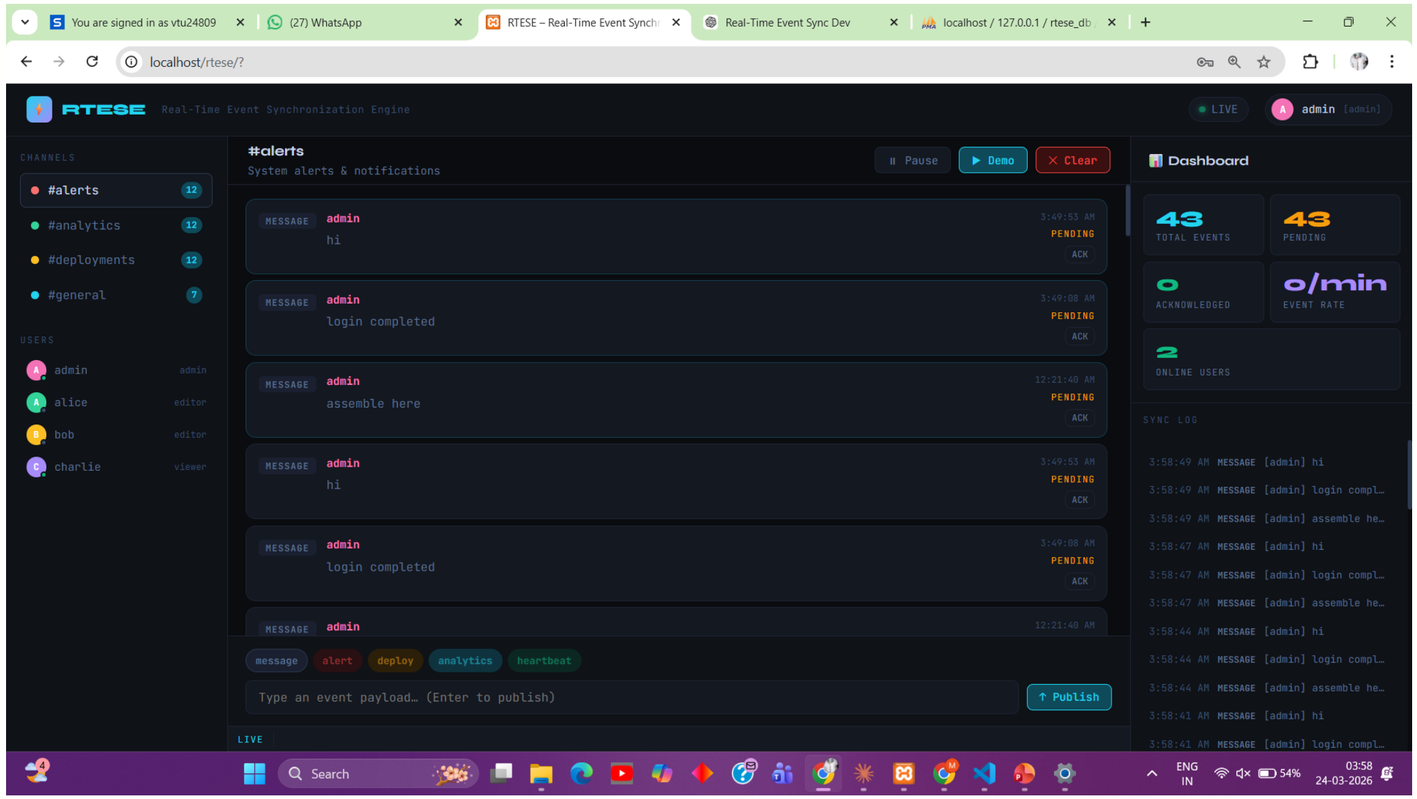
1. Login Page



- This page authenticates users using credentials stored in the database.
- It ensures secure access based on user roles.

IMPLEMENTATION SCREENSHOTS

2. Main Module (Dashboard)



- The dashboard is the central component for real-time event synchronization.
- It allows users to publish, view, and manage events dynamically.

ADVANTAGES & LIMITATIONS

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Advantages

- Real-time event updates improve system responsiveness
- Channel-based organization makes data easy to manage
- User-friendly interface for better usability
- Supports multiple users with role-based access
- Efficient data handling using MySQL and PHP backend

Limitations

- Uses polling instead of true real-time WebSockets
- Performance may reduce with large data volume
- Basic security (no encryption or hashing implemented)
- Limited scalability for large distributed systems
- Requires continuous server availability (XAMPP)

Thank you